

Wave Release Coordination under Vertical Resource Constraints in Multi-Story AMR Warehouse

Abstract

In multi-story fulfillment warehouses, autonomous mobile robots (AMRs) are confined to individual floors and share a few freight elevators, so elevator capacity is the primary bottleneck. The first tactical decision, wave composition (which orders to release together), largely fixes the makespan before any AMR moves. However, prior work has not treated it as a decision variable coupled to shared vertical transport. We formalize this gap as the wave-release coordination problem under vertical resource constraints and address it with a two-stage scheduler: a tactical stage composes each wave, and an operational stage dispatches the AMRs that deliver its orders through the shared elevators. We hold the operational stage fixed and study the tactical stage with two analytical tools. Both tools build on a three-dimensional representation $\Phi = (C, I, T)$ of each wave: its vertical spread, directional imbalance, and temporal clustering. The first, the Bound-and-Gap decomposition, is diagnostic: it splits the value of wave composition into a structural ceiling H_{up} that no policy on Φ can recover and a recoverable slack M_Φ , and identifies M_Φ as a standard predict-then-optimize loss that existing methods can close. The second, the Model-Dominance Hedge Rule, is prescriptive: a closed-form rule that selects which wave-structure to release without identifying the true elevator model. Under a conditional dominance result that we establish, this choice is provably the distributionally-robust optimum. The two tools act on the same representation and are complementary: the Bound-and-Gap decomposition measures what a wave-structure choice is worth and how much of it is reachable, while the Hedge Rule selects the robust one. A pre-registered study of 106,800 simulations reports each tool’s outcome as it stands, and the two tools’ results are different: the Hedge Rule passes every gate (per-wave dominance 99.2%, matching the distributionally-robust optimum), while the Bound-and-Gap identities hold exactly (72/72) and its resolution gate, met in 79%, fails only where the gap is genuinely small. The binding limitation on makespan is therefore predominantly capacity-side (the structural ceiling about 72% of wave-design value), and the tools’ value is to identify, regime by regime, when wave composition is worth pursuing and when the lever is capacity instead.

Introduction

The growth of e-commerce and the demand for next-day delivery have driven distribution centers into dense urban areas, where land is scarce and horizontal expansion is constrained. In response, operators increasingly adopt multi-story fulfillment buildings: floors stacked vertically, with autonomous mobile robots (AMRs) handling horizontal movement on each floor and a few freight elevators connecting them. As a result, the primary bottleneck has shifted from floor space and fleet size to elevator capacity, with AMRs frequently idling as they wait for an elevator.

Operators relieve this congestion by releasing orders in waves: within a short release window, a selected subset of orders is dispatched simultaneously. The composition of each wave, that is, which orders to bundle, directly determines how many vertical transitions the elevators must serve, in which directions, and with what temporal stagger. Therefore, it sets the difficulty of the downstream dispatch problem before any AMR moves. In current practice, wave composition is set heuristically, by

destination floor, carrier cutoff, or arrival order. However, the wave layer is the cheapest place to relieve a vertical bottleneck, because re-batching orders into different waves changes neither the layout nor the fleet size.

Four elements, namely wave composition, multi-story deployment, flexible AMR fleets, and shared-elevator capacity, have largely been studied in isolation. The closest precedents treat the lift as a constraint under fixed wave content (Chakravarty et al., 2025) or batch orders inside a single vertical lift module (VLM) (Nicolas et al., 2018). Multi-story Robotic Mobile Fulfillment System (RMFS) work avoids the coupling by confining shuttles to a single floor (J. Wu et al., 2025). Planar AMR work optimizes wave cardinality on a single floor (Qin et al., 2024). Each of these studies avoids coupling by restricting the setting: fixing the wave content, or confining transport to a single vehicle or a single floor. In the general setting we study, however, a flexible AMR fleet shares building-wide elevators and wave composition is itself a decision. Here the coupling cannot be separated: wave composition fixes the elevator trip distribution (how many vertical transitions, in which directions, with what stagger) before the fleet scheduler acts, so the tactical and operational layers cannot be optimized in isolation without losing the bottleneck interaction. We formalize the intersection of these four elements as the wave-release coordination problem under vertical resource constraints. Section 2 details the precedents.

The problem decomposes naturally into two stages: a tactical stage composes each wave, and an operational stage dispatches the AMRs that deliver their orders through the shared elevators. We hold the operational stage fixed and study the tactical stage with two complementary analytical tools, both built on a three-dimensional representation $\Phi = (C, I, T)$ of each wave: its vertical spread C , directional imbalance I , and temporal clustering T . Two structural features motivate the tools. First, the value that wave structure can deliver varies across operating scenarios. The shared elevator is the bottleneck: in some scenarios makespan is controlled by the elevator’s capacity and can be relieved only by adding capacity, while in others it depends on how waves are composed, so the wave-release policy can relieve it. The value therefore has two parts. One is fixed by capacity, and no wave-release policy can recover it. The other is slack that a Φ -informed wave-release policy can recover, the operational dispatch being held fixed. Which part dominates shifts across scenarios, so the value of composing waves must be measured before it is pursued. Second, the operator does not know which elevator model is true, but on almost every wave the candidate models rank the makespans in the same order. Among the candidate models the most conservative, true co-occupancy batching, is weakly larger than the others, hence the slowest, and serves as a worst-case reference. This ordering holds wave by wave, not only on average. The operator can therefore score the candidate waves against that single reference and release the one that minimizes makespan under it. That choice is robust: we prove it is the distributionally-robust optimum, with a computable worst-case-loss bound. The two tools act on the same decision (which wave-structure to release): the first measures how much that choice is worth and how much of it a wave-release policy can recover, and the second uses the reference model to select the robust one.

This paper makes three contributions.

(C1) Problem formulation. We formalize the wave-release coordination problem under vertical resource constraints as a two-stage scheduler in which a tactical stage composes waves and an operational stage delivers the released orders through shared elevators, the two stages coupled through

shared elevator capacity. Prior work mostly avoids this coupling by simplifying the setting. Our formulation models it explicitly.

(C2) Methodology. We develop two analytical tools on the structured representation $\Phi = (C, I, T)$. The first, the Bound-and-Gap decomposition, splits the value of wave design into a structural ceiling H_{up} that no policy operating on Φ can recover and a recoverable slack M_Φ . we prove (Theorem 1) that M_Φ is exactly a predict-then-optimize decision loss, the object studied as Smart-Predict-then-Optimize regret (Elmachtoub & Grigas, 2020), so it can be recovered with existing methods. The second, the Model-Dominance Hedge Rule is a closed-form rule that selects which wave-structure to release without identifying the true elevator model. We prove (Theorem 2) that, under a conditional per-wave dominance result we establish (Proposition 2), this choice coincides exactly with the Wasserstein distributionally robust optimum (Mohajerin Esfahani & Kuhn, 2018) and carries a computable worst-case-loss bound. The two tools act on the same representation and are complementary: the decomposition measures what a wave-structure choice is worth and how much is recoverable, while the Hedge Rule selects the robust one.

(C3) Validated insights. We evaluate both tools in a pre-registered, large-scale simulation study (106,800 simulations across 18 configurations), and report each tool’s outcome as it stands. Both results are different. The Model-Dominance Hedge Rule passes every pre-registered gate: per-wave dominance holds in 99.2% of waves (96.0% in the worst cell) and across per-trip capacities $c \in \{2, 3, 4, 5\}$, and its chosen structure matches the Wasserstein-DRO optimum in every configuration. The Bound-and-Gap identities hold exactly in all 72 sub-cells, with the capacity-side ceiling H_{up} accounting for about 72% of the mean value and dominating the smaller recoverable slack M_Φ ; the statistical-resolution gate is met in 79% of sub-cells (a partial pass, one sub-cell short), and the unresolved sub-cells are those where the value is genuinely small, so the decomposition resolves wave-design value wherever that value is substantial. For these warehouses, the binding limitation on makespan is therefore predominantly capacity-side. This is precisely the tools’ managerial payoff: they tell an operator, scenario by scenario, when smarter wave composition is worth the effort and when the lever is instead capacity or a finer partition.

Section 2 reviews the related literature. Section 3 formulates the problem. Section 4 develops the two tools. Section 5 presents computational experiments. Section 6 concludes.

Related works

The wave-elevator coupling we study sits between several established bodies of work, each touching on a piece of our setting without addressing the joint problem. We position our contribution against the literature in three steps: first we summarize the broader warehouse-OR landscape through three recent surveys; then we walk through the four problem-class threads whose intersection defines our gap (multi-story AMR and elevator group control, wave and order batching as a tactical decision, tier-captive multi-story RMFS, and data-driven fleet coordination), reading each thread for what it supplies and what it omits relative to our setting. Finally, we summarize three methodological strands on which our two analytical tools rest, with precise equivalences established in Section 4.

Three surveys frame the landscape. Boysen et al. (2019) survey warehousing in the e-commerce era, identifying the storage-and-retrieval challenges that motivate vertically stacked fulfilment buildings; Azadeh, De Koster, et al. (2019) review robotized, and automated warehouse systems with

a dedicated treatment of robotic mobile fulfilment; and Pardo et al. (2024) provide the most recent taxonomy of the order-batching family of problems, classifying decisions by their strategic, tactical, or operational layer. Within the scope of these surveys, we have not found the multi-story wave-elevator coupling treated as a standalone problem class, consistent with our positioning of this work as a formalization step.

The physical infrastructure of our setting, floors connected by shared freight elevators, has been studied piecewise in the literature we surveyed, with the pieces rarely brought together. Multi-agent robot-lift scheduling on a fixed task set is the closest precedent at the elevator layer: Chakravarty et al. (2025) compute provably optimal lift schedules for AMR fleets via Boolean satisfiability, achieving exact optimality on hotel and hospital deployments. Their formulation takes the robot-task assignment as exogenous and finds the time-optimal schedule on a fixed task set; in our setting, that fix is what we must decide, because the composition of the task set determines downstream elevator concurrency and is the dominant tactical lever for makespan. We treat wave composition as an upstream decision variable that generates the task set their methods would then optimally schedule, with the two layers coupled through shared elevator capacity. The elevator subsystem itself has a parallel literature in elevator group control: Crites et al. (1998) remain the canonical reference for learned dispatch policies under stochastic passenger arrivals, and Tsai et al. (2025) extend the family to standby strategies in smart buildings. These works optimize the elevator dispatch given an exogenous arrival process; we optimize that process by choosing the wave that generates it. At the storage layer, Azadeh, Roy, et al. (2019) analyze vertical robotic storage and retrieval systems, but the vertical degree of freedom there is robot-internal rather than building-shared. Multi-story robotic mobile fulfilment provides a complementary thread: Z. Wu et al. (2024) address inbound job scheduling in four-way shuttle systems, and Tadumadze et al. (2023) study order and pod assignment in multi-level RMFS. Both adopt a tier-captive transport model in which each shuttle or pod is assigned to a specific layer, so layer-level operations decouple, and wave composition does not appear as a decision variable. In our setting, by contrast, flexible AMRs queue at a shared elevator pool, so orders released to one floor consume vertical capacity that orders on other floors are competing for; the layers therefore couple through the shared elevator, and a layer-by-layer decomposition is no longer adequate.

Treating the composition of an order batch as a decision variable, not merely its size, has a long-standing tradition in warehouse operations research. The studies we reviewed in this area primarily focus on single-floor scenarios. Gademann et al. (2001) introduced wave picking in parallel-aisle warehouses with a makespan objective, the closest precedent to ours in objective function; Bozer & Kile (2008) and the canonical treatment of Bartholdi & Hackman (2019) embed the decision in walk-and-pick systems. Ardjmand et al. (2018) minimize order-picking makespan with multiple pickers under wave picking, an objective identical to ours, but on a single floor without a shared vertical resource. Rasmi et al. (2022) extend wave picking to the mixed-shelves storage strategy in e-commerce warehouses; Schiffer et al. (2022) derive optimal picking policies in the same e-commerce setting; Haouassi et al. (2022) integrate order line batching with batch scheduling and picker routing under split-order flexibility. Modern AMR-assisted variants enlarge the decision class: Scholz et al. (2017) formalize the Joint Order Batching, Assignment, Sequencing, and Picker Routing Problem; Žulj et al. (2022) lift it to AMR-assisted picker-to-parts systems; Qin et al. (2024) study multi-tote AMR fulfilment and identify an empirical wave-size optimum near 100 orders, where wave size is a

cardinality parameter rather than a composition decision. Two device-level batching threads are adjacent rather than parent: Nicolas et al. (2018) formulate order batching for vertical lift modules where each module is a single mechanical device and the decision is which order lines to retrieve in one tray load, and Boysen et al. (2018) address the minimum order spread sequencing problem in automated sorting; the mathematics in both is internal to the device and does not generalize to a multi-elevator capacity shared across a horizontal AMR fleet. The Pardo et al. (2024) taxonomy classifies all of these works at the tactical layer of order-batching decisions, and our wave-composition decision belongs to the same layer; what is new in our setting is that the tactical decision is evaluated against a shared vertical resource that serves a building-wide AMR fleet. We extend this decision class to the multi-story setting and represent wave composition through $\Phi = (C, I, T)$, where each axis captures a distinct vertical-resource stress.

A parallel research stream addresses warehouse fleet control through learning rather than formal optimization, and we summarize it here to position our contribution as complementary rather than competitive. Crites et al. (1998), already mentioned for elevator group control, established the foundational reinforcement-learning treatment of vertical-resource scheduling; the line continues today in AMR-specific work. Ma et al. (2025) propose an end-to-end deep reinforcement-learning framework for modular task allocation in autonomous mobile systems and report state-of-the-art performance on multi-robot dispatching benchmarks; Dhanaraj et al. (2025) study proactive contingency-aware task allocation and scheduling in multi-robot multi-human cells via hindsight optimization; Wen & Ma (2024) develop an indicator-based evolutionary algorithm for multi-objective multi-robot task allocation; Wesselhöft et al. (2022) survey reinforcement-learning approaches to AMR fleet control and identify six open challenges, including the sim-to-real gap and data efficiency. These approaches obtain policies without an explicit elevator-model commitment and would, in principle, sidestep the model uncertainty problem that our Hedge Rule addresses. We see two reasons the analytical route is complementary rather than dominated. First, the diagnostic value of our Bound-and-Gap decomposition does not transfer to learned policies, which optimize a single scalar objective and do not expose the partition-intrinsic-versus-recoverable split. Second, the closed-form Hedge Rule requires no training data and provides a deployment-ready reference against which any learned policy can be measured. The two paradigms, therefore, answer different questions about the same physical system.

Three threads of prior methodological work inform our analytical apparatus: the first two are connected to our tools by precise equivalences established in Section 4, and the third by a structural contrast. First, the prediction-to-decision regret literature (Chenreddy & Delage, 2024; Elmachoub & Grigas, 2020; Vera et al., 2020) bounds the algorithmic loss when a learned predictor drives a downstream optimizer. Theorem 1 in Section 4.1.2 gives this regret a partition perspective: the policy-slack component M_Φ of our decomposition equals the Smart-Predict-then-Optimize regret of the partition-constant predictor class, so the literature’s training-time regret object and our post-hoc, structured-feature decomposition refer to the same underlying quantity on a finite partition. Our contribution is the decomposition itself, which separates this SPO regret from an independent partition-intrinsic headroom term H_{up} . Second, Wasserstein distributionally robust optimization (Blanchet & Murthy, 2019; Gao & Kleywegt, 2023; Mohajerin Esfahani & Kuhn, 2018) hedges decisions against worst-case distributions within a transport-distance ambiguity ball of a nominal

model, typically reformulated as a convex program; the underlying robust-optimization framework dates to Delage & Ye (2010) and is surveyed by Bertsimas et al. (2011). Theorem 2 in Section 4.2.2 shows that under a chain of per-wave stochastic dominance conditions, our minimax wave-corner selection over $\{M_1, M_2, M_3\}$ coincides with the Wasserstein distributionally robust solution at the appropriate ambiguity radius, collapsing the convex program into a closed-form one-line rule. Our contribution is the dominance-conditional collapse, which exploits an empirically verified structural property to replace optimisation with identification. Third, robust scheduling under model uncertainty (Lu & Shen, 2021; Wiesemann et al., 2013) hedges against parametric variation within a single agreed-upon model class. Our elevator-modelling uncertainty is structurally different: throughput aggregation and true co-occupancy batching are qualitatively distinct models, not parameterizations of one family. The Model-Dominance Hedge Rule addresses this structural model-class disagreement rather than parametric uncertainty within a class.

Collectively, the four problem-class threads each focus on a different aspect of wave-elevator coupling. This paper formalizes the integration of four-way interaction, wave composition, multi-story deployment, flexible AMR fleets, and shared elevator capacity into a single decision problem. The methodological threads provide analytical grounding for our two tools: the first two supply the frameworks our equivalence theorems extend, and the third positions them against a competing approach. The empirical findings demonstrating these contributions in action are deferred to Sections 5 and 6. We formalize the joint problem class in Section 3 and develop the two analytical tools for it in Section 4.

Problem Formulation

We consider a multi-story fulfillment warehouse with F floors connected by E shared freight elevators. A fleet of $|A|$ floor-bound autonomous mobile robots (AMRs) handle horizontal travel on each floor; floor-to-floor transfers go through the elevators, with each elevator carrying at most c AMRs per trip. Orders arrive over a planning horizon, each with a known source floor, destination floor, and release time. A wave is a subset of orders released into the system simultaneously at the start of a release window of length Δ . The system performance metric is the wave makespan: the time at which the last order in the wave is delivered, given a fixed operational dispatch policy.

This setting decomposes naturally into two coupled decision layers. The tactical layer chooses which orders to bundle into a wave W , namely the wave composition decision; this is the focus of our analysis. The operational layer determines, given W , which AMR serves which order, which AMRs board each elevator trip, and in what order. We delegate these decisions to a deterministic dispatch policy, defaulting to capacity-bounded FIFO boarding under the elevator model $M \in \mathcal{M}$, with full simulator details given in Section 4. We model only the wave-composition decision as a formal variable; downstream AMR-elevator execution is a simulator-realized policy. This simplification is deliberate: the wave-elevator coupling is fully captured by $\Phi(W)$ at the partition resolution on which our analytical tools operate; finer partitions reveal additional structure within Φ -cells, a property formalized as the refinement monotonicity result of Corollary 1 in Section 4. Operational lift scheduling is well studied (Chakravarty et al., 2025); the upstream wave layer that prior work treats as exogenous is formalized in this paper.

Sets and indexes

$\mathcal{O}$: set of all orders over the planning horizon, indexed by o .

$\mathcal{F} = \{1, 2, \dots, F\}$: set of floors, indexed by f .

A : AMR fleet, indexed by a .

$\mathcal{E}$: set of elevators, indexed by e .

W_ω : sequence of waves over the planning horizon, indexed by ω .

$\mathcal{M} = \{M_1, M_2, M_3\}$: elevator-model set, indexed by m .

Parameters

All parameters below are deterministic under elevator models M1 and M2. Under M3, the timing primitives (τ_s, τ_e, τ_d) become lognormally perturbed with scale $\sigma_{\{M_3\}} \in \{0.10, 0.20\}$. Numerical values are reported in Section 5.

Warehouse and fleet:

$F \in \mathbb{Z}_{\geq 2}$: number of floors

$|A|$: AMR fleet size

E : number of elevators

$c \in \mathbb{Z}_{\geq 1}$: per-trip elevator capacity

Timing primitives:

$\tau_s \in \mathbb{R}_{>0}$: per-event AMR service time at pickup or dropoff

$\tau_e \in \mathbb{R}_{>0}$: per-floor elevator travel time

$\tau_d \in \mathbb{R}_{>0}$: elevator dwell time per stop

Operator-set release controls:

$W_{\min}, W_{\max} \in \mathbb{Z}_{\geq 1}$: wave cardinality bounds

$\tau_\omega \in \mathbb{R}_{>0}$: release window start of wave ω

$\Delta \in \mathbb{R}_{>0}$: release window length

Order attributes (input data):

$s_o \in \mathcal{F}$: source floor of order o

$d_o \in \mathcal{F}$: destination floor of order o (same-floor orders with $d_o = s_o$ are admissible; they contribute no elevator demand and are excluded from the $I(W)$ computation)

$r_o \in \mathbb{R}_{\geq 0}$: release time of order o

Decision variable

The model has a single decision variable, the tactical wave composition.

• $x_o \in \{0, 1\}, \forall o \in \mathcal{O}$: wave-inclusion indicator, with $x_o = 1$ if order o is released in the current wave.

The wave composition is the induced set $W = \{o \in \mathcal{O}: x_o = 1\}$.

Assumptions

The formulation rests on five assumptions, each stated explicitly, so the scope of validity is identifiable.

A1. Single-order AMR carriage. Each AMR transports at most one order at a time. This aligns with industry practice for parcel-scale fulfillment and rules out multi-tote AMR variants (Qin et al., 2024).

A2. Source-before-destination sequencing. For each served order o , the assigned AMR visits the source floor s_o before the destination floor d_o .

A3. Capacity-bounded FIFO boarding. AMRs join the elevator queue in arrival order and board up to c at a time under the elevator model M .

A4. Intra-floor travel collapsed into service time. AMR horizontal travel within a floor is summarized by the per-event service time τ_s , with no explicit congestion or routing on the floor.

A5. Elevator model is a fixed property of the warehouse, unknown to the operator at decision time. The elevator model $M \in \mathcal{M}$ is a structural property of the physical elevator subsystem (its boarding logic and trip-time aggregation), exogenous to the wave-composition decision. Because the warehouse OR literature uses both throughput-aggregation (Bartholdi & Hackman, 2019) and co-occupancy batching (Chakravarty et al., 2025), we do not commit to a single convention.

Structured-feature representation

We summarize each candidate wave by a three-dimensional fingerprint $\Phi(W) = (C(W), I(W), T(W))$, a function of the decision $\{x_o\}$, on which the analytical tools of Section 4 operate. The three axes each capture a distinct physical mechanism of vertical resource stress.

1. **Vertical activity diversity** $C(W) \geq 0$. Shannon entropy (natural logarithm) of the empirical floor distribution taken over the union of source and destination floors of orders in W . With

$$p_f(W) = \frac{|\{o \in W: s_o = f\}| + |\{o \in W: d_o = f\}|}{2|W|}, \text{ we define } C(W) = -\sum_{f \in \mathcal{F}} p_f(W) \ln p_f(W), \text{ with } 0 \ln 0 = 0.$$

High C indicates that orders span many floors (diversified vertical activity); low C indicates concentration on a few floors.

2. **Directional imbalance** $I(W) \in [0, 1]$: It measures how one-directional the cross-floor traffic is. With $W^\uparrow = \{o \in W: d_o > s_o\}$ and $W^\downarrow = \{o \in W: d_o < s_o\}$ (same-floor orders excluded from both), $I(W) = \frac{|W^\uparrow| - |W^\downarrow|}{|W^\uparrow| + |W^\downarrow|}$ when $|W^\uparrow| + |W^\downarrow| > 0$, and $I(W) = 0$ otherwise.

3. **Temporal clustering** $T(W) \geq 0$: It is the coefficient of variation of per-order release times, measured relative to the wave's release-window start. $T(W) = \frac{\sigma_r(W)}{\mu_r(W)}$. High T means orders arrive in tight bursts; low T means they spread evenly across the window.

The three axes capture conceptually distinct mechanisms: multi-floor spread, up-versus-down asymmetry, and temporal bunching, each with a clear physical meaning. We treat Φ as a conceptual decomposition rather than a predictive surrogate.

Elevator models

The dispatch policy is realized through a simulator under one of three elevator models:

1. **M_1 (throughput aggregation)**: Each elevator is modeled as a server with a per-AMR throughput rate that incorporates batched-trip efficiency into a single multiplier.
2. **M_2 (true co-occupancy batching)**: At each elevator, the next c AMRs in the queue board together and ride as a unit, with realistic per-trip capacity, dwell time, and direction handling.
3. **M_3 (stochastic batching)**: It extends M_2 with lognormal noise on per-trip duration at scale σ_{M_3} , used for robustness sensitivity.

M_1 and M_2 are both well-attested in the warehouse OR literature; the structural disagreement between throughput aggregation and explicit co-occupancy is the methodological gap our Hedge Rule resolves.

Objective function

The objective function minimizes the expected wave makespan under the chosen elevator model:

$$\min_{W \in \mathcal{O}} \mathbb{E}[C_{max}(W; M, \xi)] \quad (1)$$

where $C_{max}(W; M, \xi) := \max_{o \in W} t_o^{deliver}(W; M, \xi)$ is the wave makespan, $t_o^{deliver}(W; M, \xi)$ is the simulator-realized delivery time of order o under elevator model M , and ξ denotes a realization of the operational randomness in the dispatch simulator. Under M_1 and M_2 , ξ is degenerate and (1) reduces to the deterministic makespan $C_{max}(W; M)$; under M_3 , ξ equals the lognormal perturbations on (τ_s, τ_e, τ_d) at scale $\sigma_{\{M_3\}}$ and the expectation is non-degenerate. The expectation in (1) is taken over ξ . We adopt the makespan rather than the average delivery time because wave throughput in multi-story fulfillment is gated by the slowest order: the next wave cannot release until elevator capacity is freed by the current wave's completion.

Constraints

The wave composition W is subject to two classes of constraints:

$$W_{min} \leq |W| \leq W_{max} \quad (2)$$

$$r_o \in [\tau_\omega, \tau_\omega + \Delta], \forall o \in W \quad (3)$$

$$\sum_{\omega} 1[o \in W_\omega] = 1, \forall o \in \mathcal{O} \quad (4)$$

$$x_o \in \{0, 1\}, \forall o \in \mathcal{O} \quad (5)$$

Constraints (2)-(5) define the wave-composition decision space: cardinality bounds, temporal feasibility, exactly-one-wave-per-order, and the binary inclusion indicator. The elevator capacity bound ($n_{aboard}(e, t) \leq c$) and the AMR fleet limit $|A|$ are not formal optimization constraints at the wave-composition stage. They are invariants enforced by the operational simulator under the dispatch policy of Assumptions A1–A3, where the wave makespan $C_{max}(W; M)$ is realized. The capacity bound enters the formulation through its effect on C_{max} (a wave with composition incompatible with c manifests as a higher $C_{max}(W; M)$ rather than as infeasibility), making it natural to absorb into the objective rather than to state as a constraint. We solve (1)-(5) one wave at a time: at each release window, the decision is over orders not yet released, so (4) is automatically satisfied. The decision object in the rest of the paper is therefore a single wave's inclusion vector.

Equations (1)-(5), together with Assumptions A1-A5, define the wave release coordination problem under vertical resource constraints. This minimal formulation is deliberate: the wave-elevator coupling is fully captured by $\Phi(W) = (C(W), I(W), T(W))$ at the partition resolution on which our analytical tools operate. Operational lift scheduling is well studied (Chakravarty et al., 2025); the upstream wave layer, which prior work treats as exogenous, is formalized in this paper. Direct solution by enumeration over $\{x_o\}_o$ is intractable in any case (combinatorial $|\mathcal{O}|$, simulator-realized objective), so Section 4's tools are how the formulation becomes operational.

- Ardjmand, E., Shakeri, H., Singh, M., & Sanei Bajgiran, O. (2018). Minimizing order picking makespan with multiple pickers in a wave picking warehouse. *International Journal of Production Economics*, 206, 169–183. <https://doi.org/10.1016/j.ijpe.2018.10.001>
- Azadeh, K., De Koster, R., & Roy, D. (2019). Robotized and automated warehouse systems: Review and recent developments. In *Transportation Science* (Vol. 53, Number 4, pp. 917–945). INFORMS Inst.for Operations Res.and the Management Sciences. <https://doi.org/10.1287/trsc.2018.0873>
- Azadeh, K., Roy, D., & De Koster, R. (2019). Design, modeling, and analysis of vertical robotic storage and retrieval systems. *Transportation Science*, 53(5), 1213–1234. <https://doi.org/10.1287/trsc.2018.0883>
- Bartholdi, J. J., & Hackman, S. T. (2019). *Warehouse & Distribution Science*. www.warehouse-science.com
- Bertsimas, D., Brown, D. B., & Caramanis, C. (2011). Theory and applications of robust optimization. In *SIAM Review* (Vol. 53, Number 3, pp. 464–501). <https://doi.org/10.1137/080734510>
- Blanchet, J., & Murthy, K. (2019). Quantifying Distributional Model Risk via Optimal Transport MATHEMATICS OF OPERATIONS RESEARCH Quantifying Distributional Model Risk via Optimal Transport. *Source: Mathematics of Operations Research*, 44(2), 565–600. <https://doi.org/10.2307/27287849>
- Boysen, N., de Koster, R., & Weidinger, F. (2019). Warehousing in the e-commerce era: A survey. In *European Journal of Operational Research* (Vol. 277, Number 2, pp. 396–411). Elsevier B.V. <https://doi.org/10.1016/j.ejor.2018.08.023>
- Boysen, N., Fedtke, S., & Weidinger, F. (2018). Optimizing automated sorting in warehouses: The minimum order spread sequencing problem. *European Journal of Operational Research*, 270(1), 386–400. <https://doi.org/10.1016/j.ejor.2018.03.026>
- Bozer, Y. A., & Kile, J. W. (2008). Order batching in walk-and-pick order picking systems. *International Journal of Production Research*, 46(7), 1887–1909. <https://doi.org/10.1080/00207540600920850>
- Chakravarty, A., Grey, M. X., Muthugala, M. A. V. J., & Elara, R. M. (2025). Toward Optimal Multi-Agent Robot and Lift Schedules via Boolean Satisfiability. *Mathematics*, 13(18). <https://doi.org/10.3390/math13183031>
- Chenreddy, A., & Delage, E. (2024). *End-to-end Conditional Robust Optimization*. <http://arxiv.org/abs/2403.04670>
- Crites, R. H., Barto, A. G., Huhns, M., & Weiss, G. (1998). *Elevator Group Control Using Multiple Reinforcement Learning Agents* (Vol. 33).
- Delage, E., & Ye, Y. (2010). Accessibility support: If you experience accessibility issues with this file, report them here Distributionally Robust Optimization Under Moment Uncertainty with Application to Data-Driven Problems. *Operations Research*, 62(6), 1358–1376. <https://www.jstor.org/stable/40792682>
- Dhanaraj, N., Nemlekar, H., Nikolaidis, S., & Gupta, S. K. (2025). Proactive Contingency-Aware Task Allocation and Scheduling in Multi-Robot Multi-Human Cells via Hindsight Optimization. *IEEE Transactions on Automation Science and Engineering*, 22, 13046–13060. <https://doi.org/10.1109/TASE.2025.3546281>

- Elmachtoub, A. N., & Grigas, P. (2020). *Smart “Predict, then Optimize.”* <http://arxiv.org/abs/1710.08005>
- Gademann, A. J. R. M. noud, Van Den Berg, J. P., & Van Der Hoff, H. H. (2001). An order batching algorithm for wave picking in a parallel-aisle warehouse. *IIE Transactions (Institute of Industrial Engineers)*, 33(5), 385–398. <https://doi.org/10.1080/07408170108936837>
- Gao, R., & Kleywegt, A. (2023). Distributionally Robust Stochastic Optimization with Wasserstein Distance. *Mathematics of Operations Research*, 48(2), 603–655. <https://doi.org/10.1287/moor.2022.1275>
- Haouassi, M., Kergosien, Y., Mendoza, J. E., & Rousseau, L. M. (2022). The integrated orderline batching, batch scheduling, and picker routing problem with multiple pickers: the benefits of splitting customer orders. *Flexible Services and Manufacturing Journal*, 34(3), 614–645. <https://doi.org/10.1007/s10696-021-09425-8>
- Lu, M., & Shen, Z. J. M. (2021). A Review of Robust Operations Management under Model Uncertainty. *Production and Operations Management*, 30(6), 1927–1943. <https://doi.org/10.1111/poms.13239>
- Ma, S., Ruan, J., Du, Y., Bucknall, R., & Liu, Y. (2025). An End-To-End Deep Reinforcement Learning Based Modular Task Allocation Framework for Autonomous Mobile Systems. *IEEE Transactions on Automation Science and Engineering*, 22, 1519–1533. <https://doi.org/10.1109/TASE.2024.3367237>
- Mohajerin Esfahani, P., & Kuhn, D. (2018). Data-driven distributionally robust optimization using the Wasserstein metric: performance guarantees and tractable reformulations. *Mathematical Programming*, 171(1–2), 115–166. <https://doi.org/10.1007/s10107-017-1172-1>
- Nicolas, L., Yannick, F., & Ramzi, H. (2018). Order batching in an automated warehouse with several vertical lift modules: Optimization and experiments with real data. *European Journal of Operational Research*, 267(3), 958–976. <https://doi.org/10.1016/j.ejor.2017.12.037>
- Pardo, Gil-Borrás, Alonso-Ayuso, & Duarte. (2024). Order batching problems: Taxonomy and literature review. *European Journal of Operational Research*, 313(1), 1–24. <https://doi.org/10.13039/501100011033>
- Qin, Z., Kang, Y., & Yang, P. (2024). Making better order fulfillment in multi-tote storage and retrieval autonomous mobile robot systems. *Transportation Research Part E: Logistics and Transportation Review*, 192. <https://doi.org/10.1016/j.tre.2024.103752>
- Rasmi, S. A. B., Wang, Y., & Charkhgard, H. (2022). Wave order picking under the mixed-shelves storage strategy: A solution method and advantages. *Computers and Operations Research*, 137. <https://doi.org/10.1016/j.cor.2021.105556>
- Schiffer, M., Boysen, N., Klein, P. S., Laporte, G., & Pavone, M. (2022). Optimal Picking Policies in E-Commerce Warehouses. *Management Science*, 68(10), 7497–7517. <https://doi.org/10.1287/mnsc.2021.4275>
- Scholz, A., Schubert, D., & Wäscher, G. (2017). Order picking with multiple pickers and due dates – Simultaneous solution of Order Batching, Batch Assignment and Sequencing, and Picker Routing Problems. *European Journal of Operational Research*, 263(2), 461–478. <https://doi.org/10.1016/j.ejor.2017.04.038>
- Tadumadze, G., Wenzel, J., Emde, S., Weidinger, F., & Elbert, R. (2023). Assigning orders and pods

- to picking stations in a multi-level robotic mobile fulfillment system. *Flexible Services and Manufacturing Journal*, 35(4), 1038–1075. <https://doi.org/10.1007/s10696-023-09491-0>
- Tsai, I. N., Wu, Y. X., Huang, Y. H., Chen, Y. C., & Ding, J. J. (2025). Optimization of Elevator Standby Scheduling Strategy in Smart Buildings. *Applied System Innovation*, 8(5). <https://doi.org/10.3390/asi8050132>
- Vera, A., Banerjee, S., & Gurvich, I. (2020). *Online Allocation and Pricing: Constant Regret via Bellman Inequalities*. <http://arxiv.org/abs/1906.06361>
- Wen, C., & Ma, H. (2024). An indicator-based evolutionary algorithm with adaptive archive update cycle for multi-objective multi-robot task allocation. *Neurocomputing*, 593. <https://doi.org/10.1016/j.neucom.2024.127836>
- Wesselhöft, M., Hinckeldeyn, J., & Kreutzfeldt, J. (2022). Controlling Fleets of Autonomous Mobile Robots with Reinforcement Learning: A Brief Survey. *Robotics*, 11(5). <https://doi.org/10.3390/robotics11050085>
- Wiesemann, W., Kuhn, D., & Rustem, B. (2013). Robust markov decision processes. *Mathematics of Operations Research*, 38(1), 153–183. <https://doi.org/10.1287/moor.1120.0566>
- Wu, J., Yang, Z., Zhen, L., Li, W., & Ren, Y. (2025). Joint optimization of order picking and replenishment in robotic mobile fulfillment systems. *Transportation Research Part E: Logistics and Transportation Review*, 194. <https://doi.org/10.1016/j.tre.2024.103930>
- Wu, Z., Zhang, Y., Li, L., Zhang, Z., Zhao, B., Zhang, Y., & He, X. (2024). Research on Inbound Jobs' Scheduling in Four-Way-Shuttle-Based Storage System. *Processes*, 12(1). <https://doi.org/10.3390/pr12010223>
- Žulj, I., Salewski, H., Goeke, D., & Schneider, M. (2022). Order batching and batch sequencing in an AMR-assisted picker-to-parts system. *European Journal of Operational Research*, 298(1), 182–201. <https://doi.org/10.1016/j.ejor.2021.05.033>